



GLOBAL ECONOMICS
GRADE: 11TH
CATCH-UP
CRITERION C4
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SECOND
TERM
2025–2026

Learning objective: Interpret confidence intervals in context for confidence interval planning decisions.

Assessment criteria:

C4: Interpret the meaning of a X% confidence interval in context.

Criteria assessment

Each assessment criterion is evaluated across the problems in this catch-up exam. A criterion is considered passed when it is correctly activated in 9 of the 10 problems of this activity.

Problem 1

Downtown Listing Mean Estimate

A housing app samples new one-bedroom listings for early-career workers in a downtown area. A 95% confidence interval for the population mean monthly rent is:

$$\mu \in [1180, 1320] \text{ USD.}$$

A relocation agency must decide whether to market this area as “typically below 1200 USD per month.” Is that claim supported by the interval?

Problem 2

Designer Income Mean Estimate

A freelance platform studies monthly income for beginner graphic designers aged 20–29. A 95% confidence interval for the population mean monthly freelance income is:

$$\mu \in [1720, 2140] \text{ USD.}$$

A training program advertises that most beginners can expect around 2000 USD per month. Is 2000 USD a plausible benchmark for the population mean based on this interval?

Problem 3

Courier Earnings Mean Comparison

Two ride-delivery apps estimate the population mean weekly earnings of active couriers. App A reports a 95% confidence interval:

$$\mu_A \in [460, 540] \text{ USD.}$$

App B reports a 95% confidence interval:

$$\mu_B \in [520, 610] \text{ USD.}$$

A student chooses one app and asks: does the evidence suggest App B has a higher population mean weekly earning than App A?

Problem 4

Credit Card Full-Payment Rate

A personal finance survey estimates the population proportion of young card users who pay their full credit card balance each month. A 95% confidence interval is:

$$p \in [0.41, 0.53].$$

A policy analyst states: “There is a 95% chance that exactly 47% of users pay in full.” Explain what is incorrect in that statement.

Problem 5

Positive Return Proportion Estimate

An investing app tracks first-year users and estimates the population proportion who finish a month with positive return. A 95% confidence interval for that population proportion is:

$$p \in [0.57, 0.69].$$

The company considers the statement: “At least 70% of users are profitable in a typical month.” Is

this statement supported by the confidence interval?

Problem 6

Student Loan Repayment Comparison

A bank compares average monthly student loan repayment among two groups of recent graduates. Group A (public sector jobs) has a 95% confidence interval:

$$\mu_A \in [240, 290] \text{ USD.}$$

Group B (private sector jobs) has a 95% confidence interval:

$$\mu_B \in [305, 360] \text{ USD.}$$

Should the bank design separate repayment guidance for the two groups?

Problem 7

Employee Cost of Living

A startup surveys first-year employees about total monthly cost of living in a major city. For the same dataset, analysts report:

$$\mu \in [2320, 2480] \text{ USD at 90\% confidence,}$$

$$\mu \in [2280, 2520] \text{ USD at 95\% confidence.}$$

Management asks which interval should be used for budgeting decisions and why.

Problem 8

Micro-Shop Revenue Mean

An online seller cooperative estimates the population mean weekly revenue for new micro-shops. A 95% confidence interval is:

$$\mu \in [680, 790] \text{ USD.}$$

A mentor says a benchmark of 800 USD per week is a realistic average target for beginners. Is 800 USD plausible as the population mean according to this interval?

Problem 9

Subscription Spending Mean Estimate

A fintech app estimates the population mean monthly subscription spending by users aged 21–29. A 95% confidence interval is:

$$\mu \in [64, 88] \text{ USD.}$$

A budgeting coach proposes a “safe cap” of 90 USD as the typical monthly mean for this age group. Based on the interval, should 90 USD be treated as a plausible population mean?

Problem 10

Branch Commission Mean Comparison

A sales startup evaluates the population mean monthly commission for first-year representatives in two branches. Branch East reports a 95% confidence interval:

$$\mu_E \in [910, 1080] \text{ USD.}$$

Branch West reports a 95% confidence interval:

$$\mu_W \in [860, 980] \text{ USD.}$$

Leadership wants to know whether East can be considered clearly stronger in average commission, and whether training incentives should differ.